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MATHEMATICS TEST - CHAPTER 5: LINEAR EQUATIONS AND INEQUALITIES

Name:		Roll No.		Section:

Total Time: 2:30 Hour **Total Marks: 150**

Time Allowed: 150 Minutes

PAPER OBJECTIVE
SECTION-A (MARKS 150)

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

Q.1 Fill the relevant bubble for each part. All parts carry one mark.

Questions	A	B	C	D	A B C D
1. The standard form of a linear equation in one variable is	$ax^2 + b = 0$	$ax + b = -c$	$ax + by = 0$	$ax + b = 0$, $a \neq 0$	○○○●
2. The exponent of the variable in a linear equation is	1	2	0	−1	●○○○
3. Which one is a linear equation in one variable?	$2x + 3 = 0$	$xy + 3 = 0$	$2x^2 + 3 = 0$	$ax + y = 0$	●○○○
4. The equation $x^2 = 4$ is non-linear because	It has two solutions	It has no constant	Exponent of x is not 1	It has a fraction	○○●○
5. The equation $xy = 14$ is non-linear because it has	Two constants	A fraction	A negative sign	Product of variables	○○○●
6. A linear equation is also called an equation of	Degree two	Degree zero	Degree one	Degree three	○○●○
7. The solution of a linear equation is a replacement for the variable that	Makes the equation true	Makes it false	Is always zero	Is always positive	●○○○
8. A linear equation in one variable in standard form has	Two solutions	Exactly one solution	No solution	Infinite solutions	○●○○
9. Equations having the same solutions are called	Equivalent equations	Quadratic equations	Radical equations	Absolute equations	●○○○
10. $2x + 3 = 4$ and $2x = 1$ are	Radical equations	Non-linear equations	Inequalities	Equivalent equations	○○○●
11. To solve fractional linear equations, we multiply by the	HCF of denominators	Sum of denominators	LCM of denominators	Largest numerator	○○●○
12. Which property is used to solve $-2x = 2/5$?	Addition	Subtraction	Square root	Division	○○○●
13. Solving $5x = 25$ uses which property?	Addition property	Subtraction property	Distributive property	Division property	○○○●
14. Solve $-5x - 8 = -33$	5	−5	4	5/2	●○○○

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Questions	A	B	C	D	A B C D
15. Solve 3x + 8 = x + 10	3	0	1	2	○○●○
16. Solve 2x + 10 = 20	4	5/2	7	5	○○○●
17. Solve 4x + 6 = −30	−9/2	−18	−10	−9	○○○●
18. Solve x/4 − x/5 = 3/10	7	6	8	12	○●○○
19. Solve −2x + 2 = −16	−9	11	10	9	○○○●
20. Solve x/2 + x/5 = 14	22	40	20	21	○○●○
21. Solve 5x + 2 = 4x + 10	16	−8	7	8	○○○●
22. Solve 5x − 9 = 2x − 9	1	2	0	−1	○○●○
23. Solve 9x − 4 = 6x − 16	−5	−3	−4	−2	○○●○
24. Solve 4x + 6 = 46	10	5	20	11	●○○○
25. Solve 4/x − 2/x = 5	2/5	−2/5	5/2	−5/2	●○○○
26. Solve 4x + 12 = 8	−1/2	−2	1	−1	○○○●
27. Solve −2x − 4 = −10	4	3	3/2	2	○●○○
28. Solve 5x − 5 = −15	−4	−3	−2	−1	○○●○
29. Solve 2x + 15 = 23	6	4	3	5	○●○○
30. Solve 3x + 12 = 6	−1	0	−2	2	○○●○
31. Solve 3x + 1 = −2	−1	0	1	−1/2	●○○○

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Questions	A	B	C	D	A B C D
32. Solve 4x + 5 = 2x + 3	0	−1/2	1	−1	○○○●
33. Solve 3x + 9 = 21	6	2	4	8	○○●○
34. Solve 3x − 3 = 0	1	3	2	1/2	●○○○
35. Solve 6x − 5 = 4x + 9	7	14	9	8	●○○○
36. Solve x/2 + x/4 = 45	59	60	61	62	○●○○
37. Solve 2x − 12 = −14	0	−1/2	−1	−2	○○●○
38. Solve 8x + 9 = 5x + 6	0	−2	−1	−1/2	○○●○
39. Solve x/3 + x/5 = 16/3	12	10	9	11	○●○○
40. An equation with the variable under a radical sign is called	A linear equation	An absolute equation	An inequality	A radical equation	○○○●
41. √(x + 1) = 7 is an example of a	Absolute value equation	Radical equation	Linear inequality	Quadratic equation	○●○○
42. To solve a radical equation, we first	Add 1 to both sides	Take the cube	Multiply by zero	Isolate the radical term	○○○●
43. In solving a radical equation, we then	Square both sides	Cube both sides	Halve both sides	Take logs	●○○○
44. An extra root that fails the original equation is called	Real root	Double root	Extraneous root	Rational root	○○●○
45. Roots of radical equations must be	Ignored	Doubled	Squared again	Checked in the original equation	○○○●
46. If an isolated radical equals a negative number, the equation has	One solution	No real solution	Two solutions	Infinite solutions	○●○○
47. If x = y, then	$x^2 = y^2$	$x^2 < y^2$	$x^2 > y^2$	x = −y	●○○○
48. If $x^2 = y^2$, then	x = y always	x = −y always	x = 0	x may or may not equal y	○○○●

Questions	A	B	C	D	A B C D
49. The solution set of an equation with only extraneous root is	{0}	φ (empty set)	{1}	R	○●○○
50. Solve √x − 6 = 5	121	25	36	1	●○○○
51. Solve √(x + 38) = 8	25	27	26	28	○○●○
52. Solve √(3x − 5) = −10	35	φ (no solution)	5	−35	○●○○
53. Solve √(5t − 2) = √(3t + 4)	3	4	6	1	●○○○
54. Solve √(2x + 41) = 9	40	22	20	10	○○●○
55. Solve √(2x − 32) = 2	20	18	19	36	○●○○
56. Solve √x + 1 = 3	4	5	3	9	●○○○
57. Solve √x + 9 = 11	3	121	2	4	○○○●
58. Solve √(x + 79) = 9	3	1	2	−2	○○●○
59. Solve √(2x) = 4	16	2	8	4	○○●○
60. Solve √(5x − 51) = 2	22	11	13	−11	○●○○
61. Solve √(3x + 66) = 9	7	5	10	6	○●○○
62. Solve √x + 2 = 6	4	5	36	16	○○○●
63. The absolute value of x is its distance from	1 on the number line	The origin of a circle	−1 on the number line	0 on the number line	○○○●
64. The absolute value of a real number is always	Negative	Positive or zero	Negative or zero	Less than zero	○●○○
65. |x| = x if	x ≥ 0	x < 0	x ≤ 0 only	x = −1	●○○○

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Questions	A	B	C	D	A B C D
66. $\|x\| = -x$ if	$x > 0$	$x \geq 0$	$x < 0$	$x = 1$	○○●○
67. $\|-5\|$ =	5	−5	0	±5	●○○○
68. $\|9\|$ =	−9	0	9	±9	○○●○
69. $\|-3\| = -(-3)$ =	−3	3	0	9	○●○○
70. The equation $\|x\| = k$ with $k > 0$ has	One solution	Two solutions k and −k	No solution	Infinite solutions	○●○○
71. The equation $\|x\| = 0$ has	One solution x = 0	Two solutions	No solution	Infinite solutions	●○○○
72. The equation $\|x\| = k$ with $k < 0$ has	One solution	Two solutions	Infinite solutions	No solution	○○○●
73. $\|ax + b\| = c$ $(c > 0)$ is equivalent to	$ax + b = c$ only	$ax + b = -c$ only	$ax + b = c$ and $ax + b = -c$	$ax + b = c$ or $ax + b = -c$	○○○●
74. $\|ab\|$ equals	$\|a\| + \|b\|$	$\|a\| - \|b\|$	$\|a\|\|b\|$	$a\|b\|$	○○●○
75. $\|a/b\|$ $(b \neq 0)$ equals	$\|a\| - \|b\|$	$\|a\|/\|b\|$	$\|a\|\|b\|$	$a/\|b\|$ only	○●○○
76. $\|a\| + \|a\|$ equals	$2\|a\|$	$\|2a\|^2$	$\|a\|^2$	0	●○○○
77. If two expressions have the same absolute value, then they are	Always equal	Equal or opposite	Always opposite	Always zero	○●○○
78. An equation with a variable inside absolute value bars is called	Radical equation	Absolute value equation	Linear inequality	Quadratic equation	○●○○
79. Solve $\|x - 2| = 2$	$\{-1, 5\}$	$\{-4, 0\}$	$\{4\}$	$\{0, 4\}$	○○○●
80. The solution set of $\|x\| = -13$ is	∅	{13}	{−13}	{−13, 13}	●○○○
81. Solve $\|2x - 5| = 14$	$\{-9/2\}$	$\{19/2\}$	$\{-9/2, 19/2\}$	$\{9/2, 19/2\}$	○○●○
82. Solve $\|x\| = 3$	$\{-3, 3\}$	$\{0\}$	$\{-3\}$	$\{3\}$	●○○○

Questions	A	B	C	D	A B C D
83. The value of |12| is	0	−12	13	12	○○○●
84. The value of |−4| is	4	−4	0	5	●○○○
85. Solve |x − 8| = 2	{5, 11}	{−10, −6}	{10}	{6, 10}	○○○●
86. Solve |2x − 9| = 8	{1/2}	{1/2, 17/2}	{−1/2, 19/2}	{−1/2, 17/2}	○●○○
87. Solve |2x + 5| = 11	{8, −3}	{−8}	{3, −8}	{3}	○○●○
88. Solve |x| = 7	{0}	{7}	{−7, 7}	{−7}	○○●○
89. Solve |x| = 6	{0}	{−6, 6}	{−6}	{6, 7}	○●○○
90. Solve |4x + 8| = 7	{−15/4}	{−1/4}	{−15/4, −1/4}	{−1/4, 15/4}	○○●○
91. Solve |x| = 5/3	{5, 3}	{5/3}	{3/5, −3/5}	{−5/3, 5/3}	○○○●
92. The solution set of |x| = −2 is	∅	{2}	{−2}	{−2, 2}	●○○○
93. The value of |6| is	6	0	−6	7	●○○○
94. The solution set of |x| = −11 is	{11}	∅	{−11}	{−11, 11}	○●○○
95. Solve |3x + 4| = 4	{0, 8/3}	{−11/3, 1}	{0}	{−8/3, 0}	○○○●
96. Solve |x| = 9	{0}	{−9}	{−9, 9}	{9}	○○●○
97. A statement that two expressions are not equal is called	An inequality	An equation	An identity	A formula	●○○○
98. The symbol "≤" means	Greater than or equal to	Less than or equal to	Not equal to	Less than	○●○○
99. The symbol "≥" means	Greater than or equal to	Less than or equal to	Not equal to	Greater than	●○○○

Questions	A	B	C	D	A B C D
100. The symbol "≠" means	Equal to	Less than	Not equal to	Greater than	○○●○
101. Which of these is a strict inequality?	$12x > 5$	$x + 3 \neq 0$	$2y - 3 \leq 0$	$4x + 5 \geq 0$	●○○○
102. An inequality using < or > is called	Compound inequality	Equation	Equivalent inequality	Strict inequality	○○○●
103. The standard form of a linear inequality is	$ax + b < 0$, $a \neq 0$	$ax + b = 0$	$ax^2 + b < 0$	$ax + by = 0$	●○○○
104. Multiplying both sides of an inequality by a negative number	Reverses the inequality sign	Keeps the sign	Makes it an equation	Doubles the solution	●○○○
105. Dividing both sides of an inequality by a negative number	Keeps the sign	Removes the sign	Makes it an equation	Reverses the inequality sign	○○○●
106. Inequalities with the same solution set are called	Equivalent inequalities	Compound inequalities	Strict inequalities	Opposite inequalities	●○○○
107. Two inequalities joined by "and" or "or" are called	Compound inequalities	Strict inequalities	Equivalent inequalities	Simple equations	●○○○
108. A compound inequality with "or" is true if	Both inequalities are true	At least one inequality is true	Neither is true	Exactly two are true	○●○○
109. A compound inequality with "and" is true if	At least one is true	Both inequalities are true	Neither is true	One is false	○●○○
110. $-6 < 5x + 3 < 5$ combines two inequalities joined by	or	but	and	not	○○●○
111. For real numbers a and b, exactly one of $a < b$, $a = b$, $a > b$ holds. This is	Trichotomy property	Transitive property	Additive property	Multiplicative property	●○○○
112. If $a < b$ and $b < c$, then $a < c$. This is the	Trichotomy property	Additive property	Transitive property	Multiplicative property	○○●○
113. If $a < b$, then $a + c < b + c$. This is the	Transitive property	Trichotomy property	Additive property	Multiplicative property	○○●○
114. If $a < b$ and $c > 0$, then $ac < bc$. This is the	Additive property	Multiplicative property	Transitive property	Trichotomy property	○●○○
115. If $a < b$ and $c < 0$, then	$ac < bc$	$ac = bc$	$ac > bc$	$a + c = b$	○○●○
116. Is $5 \geq 1$ true?	No	Only if $5 = 1$	Cannot say	Yes, since $5 > 1$	○○○●

Questions	A	B	C	D	A B C D
117. Is $7 \le 7$ true?	No	Only if $7 < 7$	Cannot say	Yes, since $7 = 7$	○○○●
118. In $-2 < 5$, multiplying by -3 gives	$6 < -15$	$6 > -15$	$-6 > 15$	$-6 < 15$	○●○○
119. $x + 5 < 8$ and $x < 3$ are	Strict equations	Absolute inequalities	Not related	Equivalent inequalities	○○○●
120. Symbols < and > were introduced by	Newton	Euclid	Descartes	Thomas Harriot	○○○●
121. Solve $7x < 6x - 1$	$x > 1$	$x < -1$	$x < 1$	$x > -1$	○●○○
122. What should be the value of k if $x < y$ shows $kx > ky$?	$k > 0$	$k = 0$	$k < 0$	$k \ge 0$	○○●○
123. Solve $-3x + 8 \le 4$	$x \ge 4/3$	$x \le -4/3$	$x \le 4/3$	$x \ge -4/3$	●○○○
124. Solve $2x - 12 \ge -7$	$x \ge 5/2$	$x \le 5/2$	$x \ge -5/2$	$x \ge -19/2$	●○○○
125. Solve $x \ge 2$ and $x \le 5$	$x \ge 2$ or $x \le 5$	$5 \le x \le 2$	$2 \le x \le 5$	$2 < x < 5$	○○●○
126. Solve $-5x - 10 \le -2$	$x \ge 12/5$	$x \ge -8/5$	$x \le -8/5$	$x \ge 8/5$	○●○○
127. Does $y = 3$ satisfy $5y - 12 > 0$?	No	Only for $y > 4$	Cannot say	Yes	○○○●
128. Solve $4x + 2 \ge -12$	$x \ge 7/2$	$x \le -7/2$	$x \ge -7/2$	$x \ge -5/2$	○○●○
129. Solve $x \ge 3$ and $x \le 10$	$x \ge 3$ or $x \le 10$	$10 \le x \le 3$	$3 < x < 10$	$3 \le x \le 10$	○○○●
130. Solve $-3x + 6 \ge 15$	$x \ge 3$	$x \le -3$	$x \le 3$	$x \le -2$	○●○○
131. The number of natural numbers x satisfying $x < 5$ is	5	6	3	4	○○○●
132. Solve $x \ge -2$ and $x \le 4$	$-2 \le x \le 4$	$x \ge -2$ or $x \le 4$	$4 \le x \le -2$	$-2 < x < 4$	●○○○
133. Solve $5x + 8 < -9$	$x < 17/5$	$x < -1/5$	$x < -17/5$	$x < -12/5$	○○●○

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Questions	A	B	C	D	A B C D
134. Solve $x \geq -1$ and $x \leq 2$	$x \geq -1$ or $x \leq 2$	$-1 \leq x \leq 2$	$2 \leq x \leq -1$	$-1 < x < 2$	○●○○
135. Solve $8x - 5 \leq 3x - 3$	$x \leq 2/5$	$x \geq -2/5$	$x \leq -2/5$	$x \geq 2/5$	●○○○
136. Solve $2x > 4$	$x < 2$	$x > 2$	$x > -2$	$x < -2$	○●○○
137. Solve $7x + 2 > 6x - 10$	$x > -12$	$x < -12$	$x > 12$	$x < 12$	●○○○
138. Solve $2x - 3 < -15$	$x > -6$	$x < -6$	$x < 6$	$x < -9$	○●○○
139. Solve $9x - 3 \leq 6x$	$x \leq 1$	$x \leq 2$	$x \leq -1$	$x \geq -1$	●○○○
140. Solve $x \geq 1$ and $x \leq 5$	$x \geq 1$ or $x \leq 5$	$1 \leq x \leq 5$	$5 \leq x \leq 1$	$1 < x < 5$	○●○○
141. Solve $5x - 4 \leq -7$	$x \leq 3/5$	$x \leq -3/5$	$x \leq 2/5$	$x \leq -11/5$	○●○○
142. Solve $-2x + 2 \leq 0$	$x \leq 1$	$x \geq 2$	$x \geq -1$	$x \geq 1$	○○○●
143. Solve $4x + 4 > 14$	$x < 5/2$	$x > 5/2$	$x > 7/2$	$x > 9/2$	○●○○
144. Solve $-3x + 9 \leq 5$	$x \geq 7/3$	$x \geq 4/3$	$x \geq -14/3$	$x \geq -4/3$	○●○○
145. Solve $-5x - 9 > 4$	$x > -13/5$	$x > 13/5$	$x < -13/5$	$x < 1$	○○●○
146. Solve $x \geq 2$ and $x \leq 8$	$x \geq 2$ or $x \leq 8$	$8 \leq x \leq 2$	$2 \leq x \leq 8$	$2 < x < 8$	○○●○
147. Solve $x \geq 4$ and $x \leq 11$	$x \geq 4$ or $x \leq 11$	$11 \leq x \leq 4$	$4 < x < 11$	$4 \leq x \leq 11$	○○○●
148. Solve $x \geq -2$ and $x \leq 0$	$x \geq -2$ or $x \leq 0$	$0 \leq x \leq -2$	$-2 < x < 0$	$-2 \leq x \leq 0$	○○○●
149. Solve $3x - 6 > x - 9$	$x < -3/2$	$x > -3/2$	$x > -1/2$	$x > 3/2$	○●○○
150. Solve $2x + 6 \geq -3$	$x \geq 9/2$	$x \leq -9/2$	$x \geq -9/2$	$x \leq 9/2$	○○●○